



BITS Pilani

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CE F231 Fluid Mechanics

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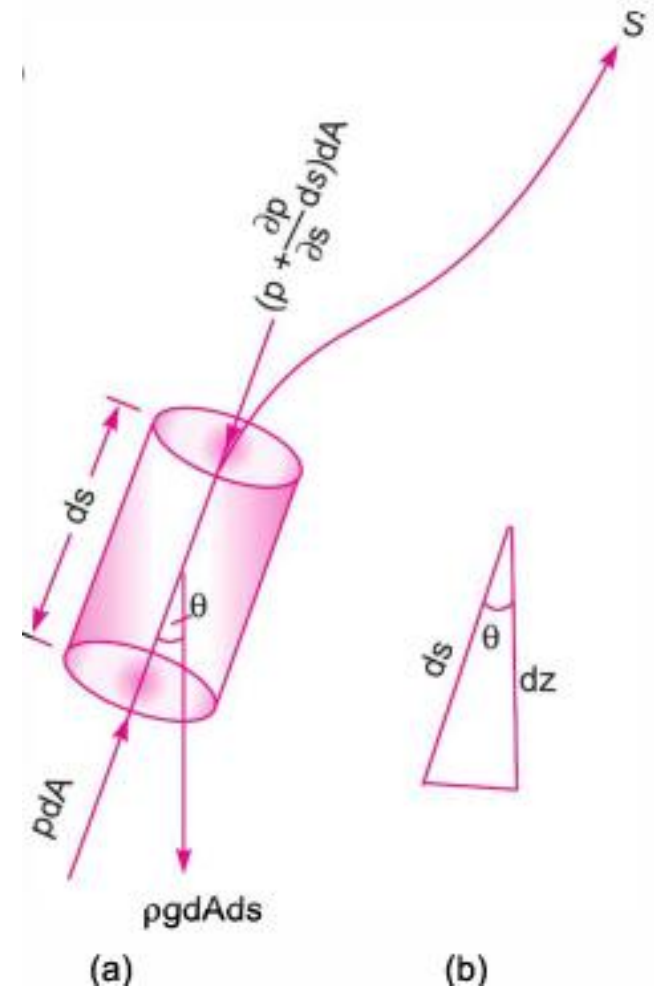
Lecture 22: Fluid Dynamics

Recap: Euler's Equation

$$\frac{dp}{\rho} + g dz + v dv = 0$$

Assumptions:

- Inviscid
- Steady flow
- Incompressible

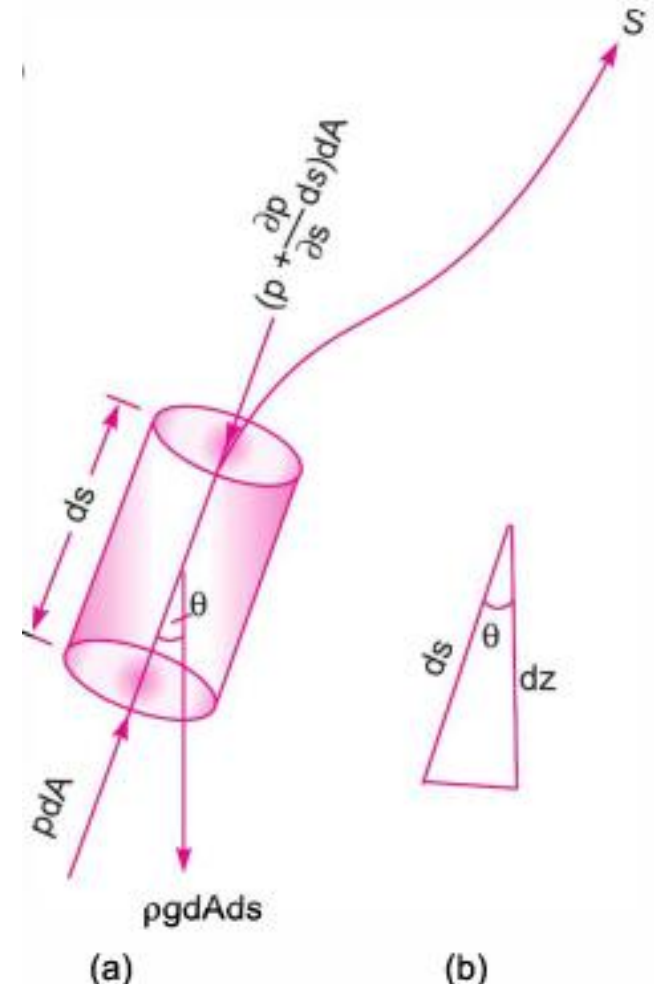


Recap: Bernoulli's Equation

$$\frac{p}{\rho g} + \frac{v^2}{2g} + z = \text{constant}$$

Assumptions:

- Inviscid
- Steady flow
- Incompressible
- Irrotational



Recap: Bernoulli's Equation

Bernoulli's equation for real fluids

Direction of flow – section 1 to section 2

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L$$

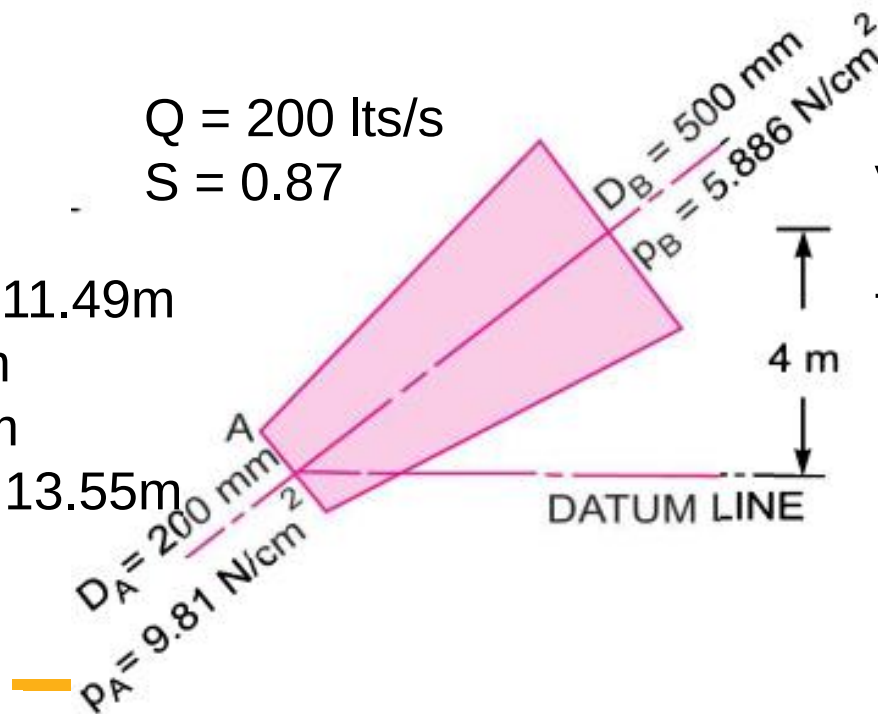
Bernoulli's Equation

Bernoulli's equation for real fluids

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2 + h_L$$

$Q = 200 \text{ lts/s}$
 $S = 0.87$

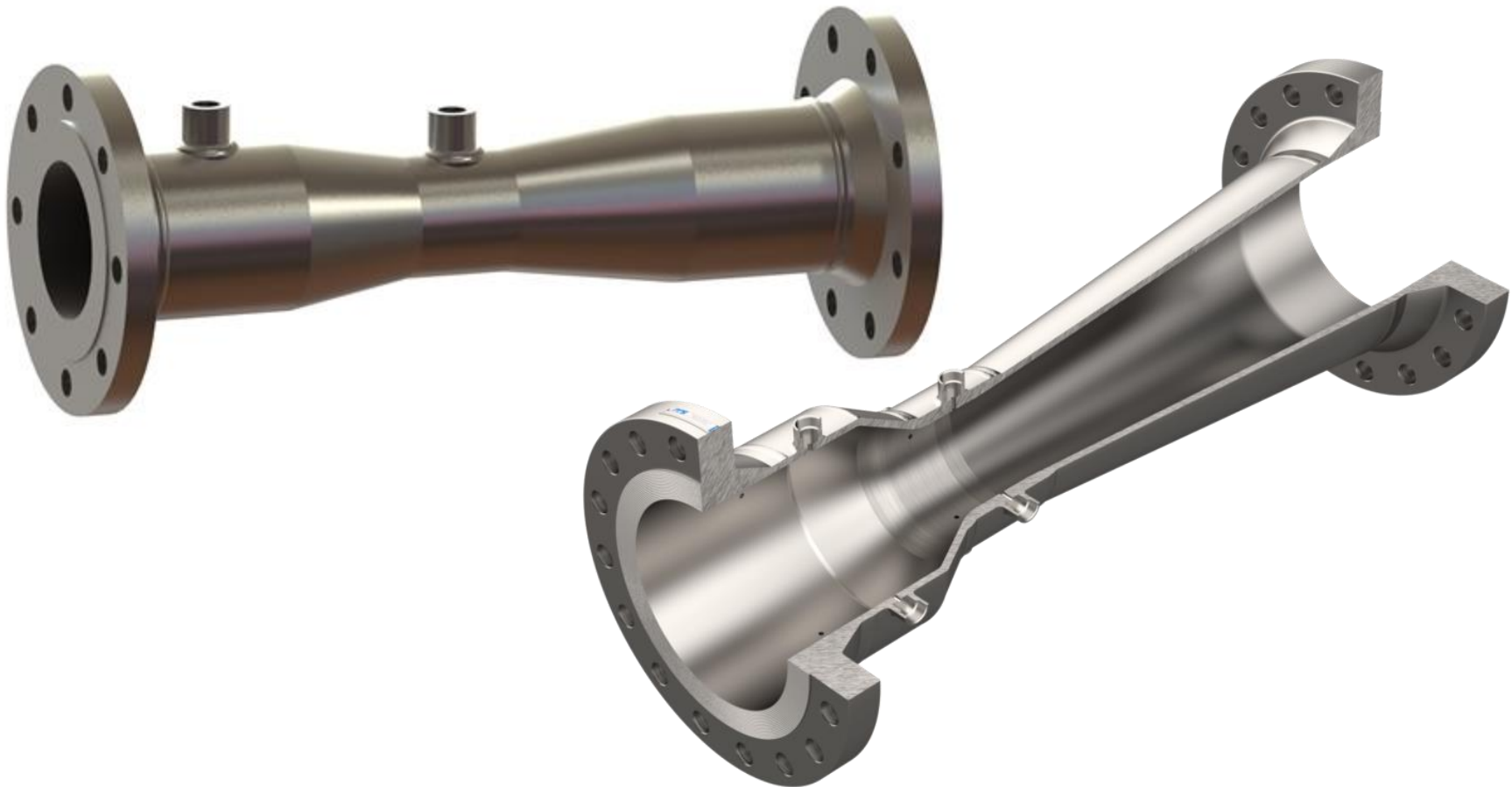
Pressure head = 11.49m
 Vel head = 2.06m
 Datum head = 0m
 Total head at A = 13.55m



Pressure head = 6.89m
 Vel head = 0.052m
 Datum head = 4m
 Total head at B = 10.94m

Venturi meter

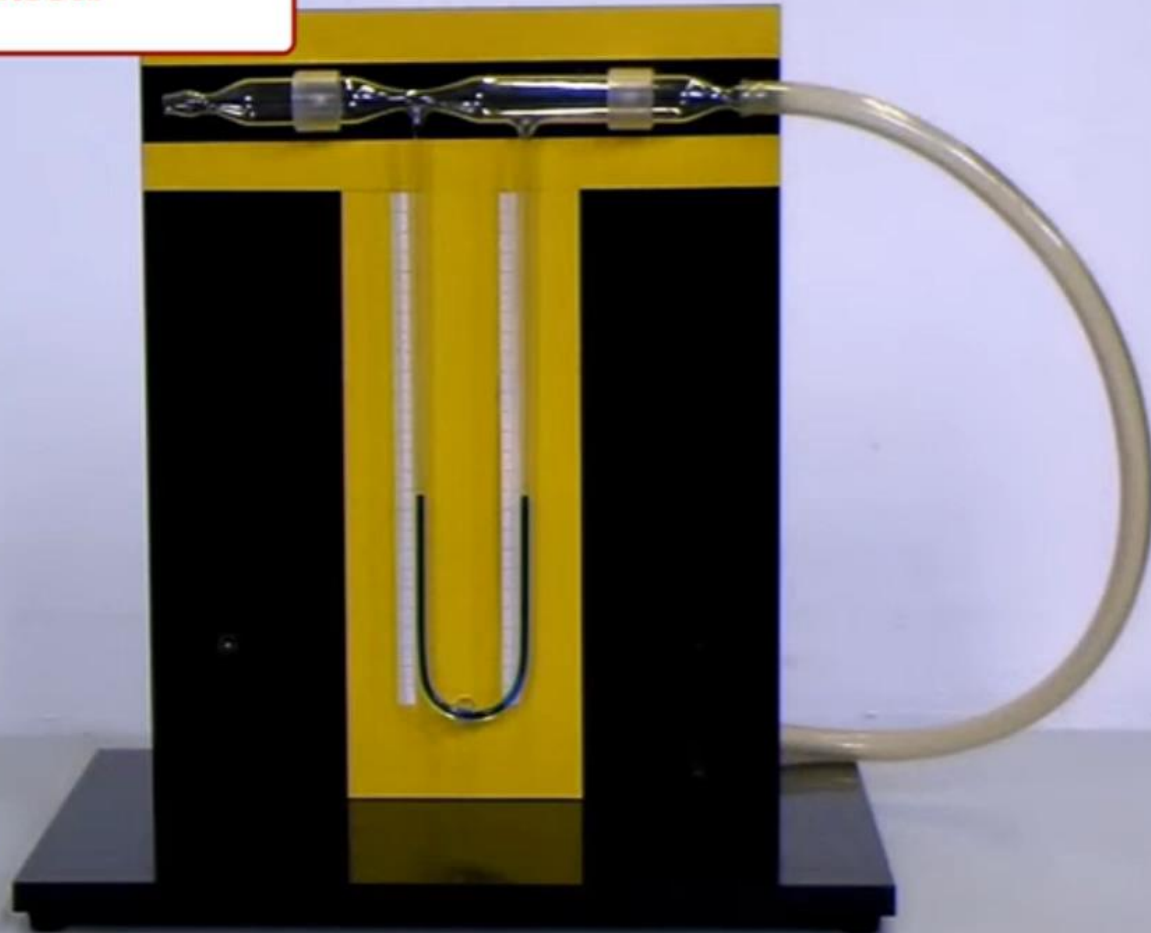
- Used for measuring rate of fluid flow through pipe



Venturi meter



Instrumentation



Venturi meter

The water is flowing through a pipe of diameter 30 cm. The pipe is inclined and a venturimeter is inserted in the pipe. The diameter of venturimeter at throat is 15 cm. The difference of pressure between the inlet and throat of the venturimeter is measured by a liquid of sp. gr. 0.8 in an U-tube manometer which gives a reading of 40 cm. The loss of head between the inlet and throat is 0.3 times the kinetic head of the pipe. Find the discharge. **[Ans. 22.64 lit./s]**



THANK YOU